

3.3.2 Respiratory diseases and treatment

Learning outcomes	Additional guidance
<i>Learners should be able to demonstrate and apply their knowledge and understanding of:</i>	
(a) the short-term and long-term effects of pollutants on the respiratory system	To include tobacco smoke, asbestos and fungal spores.
(b) the causes and symptoms of chronic bronchitis and emphysema (COPD), asthma and lung cancer	
(c) comparisons of acute and chronic diseases	
(d) the treatment of asthma	To include the use of beta agonists and steroids.
(e) the importance of plants as potential sources of medicinal drugs	To include the use of theophylline (from <i>Theobroma cacao</i>) in the treatment of COPD and asthma, topotecan (from <i>Camptotheca acuminata</i>) in the treatment of lung cancer, aspirin (from <i>Salix spp.</i>) and quinine (from <i>Cinchona spp.</i>).
(f) the design and use of clinical trials to assess the value of treatments	To include the role of each phase of a clinical trial and the importance of the recruitment and size of sample, randomisation, placebos and double-blind trials. <i>M1.5</i> HSW9, HSW10, HSW11
(g) the role of NICE (The National Institute for Health and Care Excellence) in providing guidelines for treatments.	To include its role in providing guidelines on clinical practice, health technologies and public health AND the importance of economic considerations in the production of guidelines. HSW9, HSW10, HSW11